COMSM0067: Advanced Topics in Programming Languages
Problem Sheet 12

Symbolic Execution — paths, path conditions and under-approximation

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1. Execute the following program symbolically, with x and y symbolic inputs α, β.

$$\mathsf{if}\ x < y\ \mathsf{then}\ (m := x)\ \mathsf{else}\ (m := y); \quad \mathsf{if}\ m > 10\ \mathsf{then}\ (m := m - 10)\ \mathsf{else}\ \mathsf{skip}$$

   (a) Draw the execution tree. How many leaves does it have?

   (b) For each leaf, give the path condition π and the final value of $\sigma(m)$.

   (c) Exactly one leaf is *infeasible*. Which, and why? Give the contradiction in its path condition.

   (d) For each *feasible* leaf, give concrete values of x and y that drive execution down it.

2. Now a loop. Let n be a symbolic input α, and consider

$$i := 0; \quad s := 0; \quad \mathsf{while}\ i < n\ \mathsf{do}\ \big(s := s + i;\ i := i + 1\big); \quad \mathsf{assert}\ s \geq 0.$$

   Unroll the loop **three** times.

   (a) Complete the table of *exiting* paths:

laps	path condition π	$\sigma(s)$	$\sigma(i)$
0	$\alpha \leq 0$	0	0
1			
2			
3			

   (b) For each exiting path, write the query that asks whether the assertion can *fail*, and say whether it is satisfiable.

   (c) Suppose we had stopped after unrolling *once*. Which executions would we have said nothing about?

3. Here is a program with a bug. The inputs a and b are symbolic.

$$q := 0; \quad \mathsf{while}\ a \geq b\ \mathsf{do}\ \big(a := a - b;\ q := q + 1\big); \quad \mathsf{assert}\ a \geq 0$$

   (a) Execute symbolically, unrolling twice. Give the path condition and $\sigma(a)$ for each exiting path.

   (b) Find a path whose assertion query is satisfiable, and give a model — concrete values of a and b that make the assertion fail.

   (c) The bug is not really about the assertion; it is about a missing precondition. What precondition on b should the program have, and which of your path conditions becomes unsatisfiable once you add it?

4. (*) *Over versus under.* Consider

$$\mathsf{if}\ x > 0\ \mathsf{then}\ (y := 1)\ \mathsf{else}\ (y := 2)$$

   with x symbolic.

   (a) Symbolic execution produces *two* states. Write them.

(b) Hoare logic's IF rule produces *one* postcondition covering both branches. Write the strongest such Q you can.

(c) Your Q from (b) describes a *set* of final states. Is every state in that set actually reachable? Is every state in the two symbolic states reachable?

(d) Now suppose a tool reports "y may be 3 here". Under which of the two approaches could that report be *spurious*, and under which would it necessarily indicate a real bug? Answer in one sentence each.